

6.3.5 Configuration Management Process

6.3.5.1 Purpose

The purpose of the Configuration Management Process is to establish and maintain the integrity of all identified project or process outputs (configuration items) and make them available to concerned parties throughout the system life cycle.

6.3.5.2 Outcomes

As a result of successful implementation of the Configuration Management Process:

- a) A configuration management strategy is defined.
- b) Items requiring configuration management are identified.
- c) Configuration baselines are established and maintained.
- d) Changes to configuration items are identified, evaluated, approved, implemented, and verified.
- e) The configuration of released items is controlled throughout their life cycle.
- f) The status and history of configuration items are recorded and made available for traceability, control, and audit.

NOTE: The **Software Configuration Management Process** is a specialization of this process and is included in the Software Support Process Group (see 7.2.2).

6.3.5.3 Activities and Tasks

The organization shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.3.5.3.1 Configuration Management Planning

This activity establishes the configuration management framework for the project or organizational scope.

- 6.3.5.3.1.1 Define a configuration management strategy, including

applicable standards, tools, methods, roles, responsibilities, and reporting mechanisms.

- 6.3.5.3.1.2 Identify items subject to configuration control, including system elements, software components, documentation, data, and supporting artifacts.
 - 6.3.5.3.1.3 Define configuration control procedures for baselines, change requests, audits, and status accounting.
 - 6.3.5.3.1.4 Establish configuration item naming and identification conventions, ensuring uniqueness, durability, and compliance with product sector standards.
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6.3.5.3.2 Configuration Identification

This activity identifies and records configuration items and establishes baselines.

- 6.3.5.3.2.1 Assign unique, durable identifiers or markings to configuration items in accordance with relevant standards and conventions.
 - 6.3.5.3.2.2 Describe configuration items using standardized formats and ensure traceability to design specifications, requirements, or equivalent documented descriptions.
 - 6.3.5.3.2.3 Consolidate evolving configuration states into documented **baselines** at designated times or under defined conditions (e.g., release milestones, reviews).
 - 6.3.5.3.2.4 Record baseline data including configuration states, rationale for baseline creation, and authorization details.
 - 6.3.5.3.2.5 Maintain configuration records throughout the system life cycle and archive them according to agreements, legislation, or best practices.
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6.3.5.3.3 Configuration Control

This activity manages changes to baselined configuration items.

- 6.3.5.3.3.1 Ensure proper identification, recording, evaluation, approval, incorporation, and verification of changes to configuration baselines.
- 6.3.5.3.3.2 Use Decision Management Process (6.3.3) to support change evaluation and selection of the most beneficial course of action.
- 6.3.5.3.3.3 Control incorporation of approved changes into the configuration, including appropriate verification and validation.
- 6.3.5.3.3.4 Maintain full traceability between change requests, approvals, and implemented baselines.
- 6.3.5.3.3.5 Protect configuration information against unauthorized changes through appropriate integrity and security controls.

NOTE: Configuration control involves managing the recording, retrieval, and consolidation of current and preceding configuration states to confirm information correctness, timeliness, integrity, and security.

6.3.5.3.4 Configuration Status Accounting and Audit

This activity provides visibility into the configuration state and verifies baseline integrity.

- 6.3.5.3.4.1 Record, maintain, and report configuration status information for each configuration item throughout its life cycle.
 - 6.3.5.3.4.2 Provide configuration status reports to concerned parties, including decision makers, auditors, and implementers.
 - 6.3.5.3.4.3 Perform configuration audits to verify that the as-built configuration conforms to the approved baseline (e.g., drawings, specifications, interfaces).
 - 6.3.5.3.4.4 Use Review (7.2.6) and Audit (7.2.7) processes to assess configuration integrity and compliance.
 - 6.3.5.3.4.5 Retain configuration audit records for traceability and continuous improvement.
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Cross-References

- **Decision Management (6.3.3)** — for evaluating and approving configuration changes.
- **Measurement (6.3.2)** — for monitoring CM performance indicators.
- **Verification (7.2.4), Review (7.2.6), Audit (7.2.7)** — for configuration validation and control.
- **Software Configuration Management (7.2.2)** — specialization for software items.